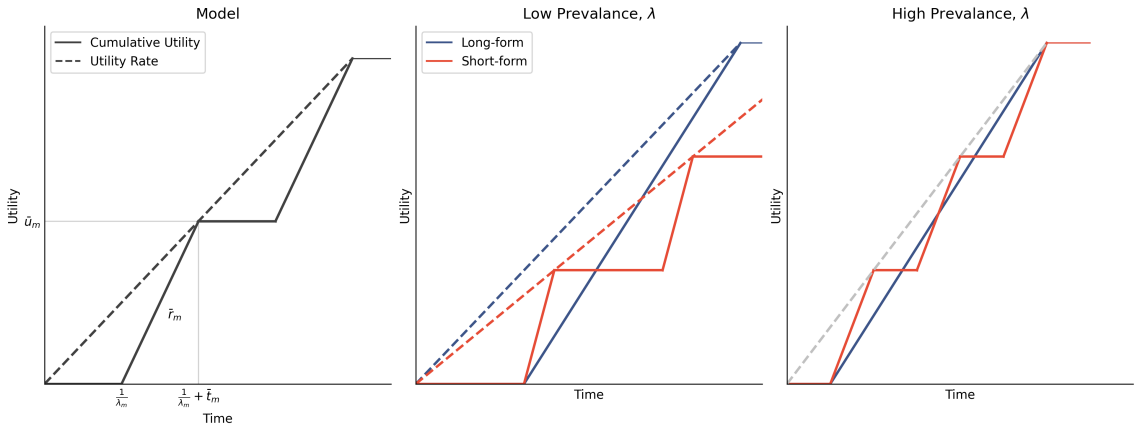

Figure 5: The media patch model. a) The expected utility rate of a media patch (dashed line) is determined by the time spent searching for (horizontal solid line) and consuming (diagonal solid line) information items. b) In a low prevalence environment long-form media has an advantage, although at low prevalence foragers are not very selective. c) At high prevalence less time is spent searching between item acquisition. To reach the same overall patch utility rate (dotted grey line), short-form media needs a higher information utility rate (gradient of the solid diagonal red line) than long-form media (gradient of the solid diagonal blue line).

Social Media

Inequality 12 includes a weaker condition for diet inclusion, $\frac{1}{\lambda_m \bar{u}_m} \leq \frac{1}{R_{env}}$. This indicates that information prevalence directly limits the minimal average size of information for diet inclusion. As information prevalence increases, foragers will tolerate media platforms with smaller and smaller information item sizes (Figure 6). More intuitively, Twitter only works in a world with instant messages — few people would go to a library in order to check out a single Tweet.

Finally, our model quantifies the selective forces acting to make media platforms more accessible. If a media platform reduces the expected search time between information encounters, $\frac{1}{\lambda_m}$, then they reduce the left hand side of Inequality 12 and become more competitive. This asymmetrically effects utility for short-form media, $\frac{1}{\lambda_m \bar{u}_m}$; for long-form media this term is already small. This could be an explanation for innovations towards minimising time spent searching in short-form media platforms such as infinite scroll and autoplay videos.

Discussion

We provide evidence that the word entropy of American English has increased over the 20th century. Furthermore, this change is marked by differences across different media categories, with the highest entropy levels found in the shortest media forms. Using a model of the attention economy based on information foraging, we show how a simple model of information selection can drive the observed changes. The attention economy model explains two results: a rise in entropy as information becomes more abundant and a rise in preferences for information dense short-form media.

Our findings offer an interesting contrast to the Linguistic Niche Hypothesis [41], which predicts a loss of complex morphological forms in English due to the influence of second language learners. There is ample evidence that English is undergoing morphological simplification [42, 43, 44], and we might expect this to be associated with a decrease in word entropy (further explored in the Supplementary Information). Our findings show the opposite. Our claim is that the pressure towards information density overcomes the effect of reduced word entropy through linguistic simplification. However it may be that a reduction in morphological complexity and a rise in information entropy are related — in attention markets people may be attracted to both simplicity [45] and novelty. Specifically, a loss in morphological complexity may be driven by a pressure towards simplicity and a reduction in the repetition of more difficult to process linguistic forms. That is, the features of the attention economy that drive rising entropy may also drive reduced morphological complexity.

Language evolution has been shown to follow a number of principles governed by human psychology.

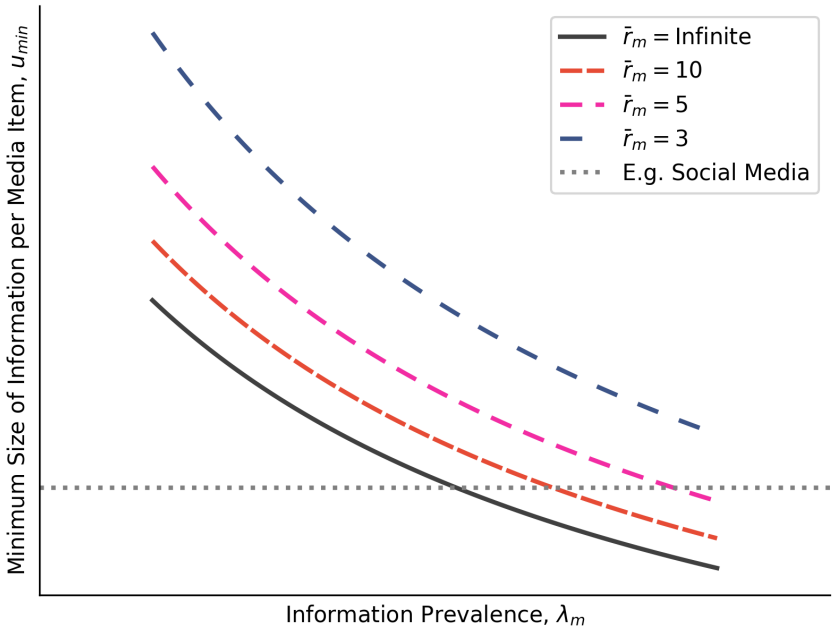

Figure 6: Minimum average information size, u_{min}, for media platform diet inclusion for varying levels of information prevalence, λ_m. Increasing average information utility rates, $\bar{r}_m$, can increase this limit only to a point. Very short-form media platforms like social media can only capture attention in a world with high information prevalence.

These principles have, for example, included features of biological and cultural evolution [46, 47], learning [45, 47, 41], cooling by expansion [48], word formation and distribution [39], and the decay of morphological complexity [41, 43]. Our results extend the psychological consequences on language evolution to word entropy in response to information abundance.

Considering people as information foragers, our model describes observed empirical changes in word entropy of English over time and both within and between media categories in response to increasing information abundance. Empirical findings support the idea that people's attention is attracted to high entropy and high complexity information [49, 50]. Our analysis of historical data shows the entropy of information markets respond predictably to increased competition. The attention economy model offers a simple explanation: humans are, within limits, information rate maximisers responding to rising information abundance and media producers adapt their content to compete for more limited attention.

Humans choices are based on more than entropy. For example, humans respond to social cues and risk [5] just as animals consider factors other than calorie rate such as macro-nutrient content and predators when foraging for food [36]. Moreover, information producers are not only interested in capturing attention, but also in influence [51, 7]. Nonetheless, just as animal foraging models have been shown to predict human behaviour in a variety of domains [52, 11, 53, 54, 55], our analyses suggests these models also extend to the shape of information evolution and cultural history, just as the co-evolutionary arguments of Darwin might have predicted [56].

Data Availability

All data generated following analysis of text samples is available at `https://github.com/chasmani/PUBLIC-the-rising-entropy-of-english-in-the-attention-economy`.

The text corpora data is not included in the public repository for copyright and size reasons. They are available:

- COHA and COCA. `https://www.corpusdata.org/`

- BNC. `http://www.natcorp.ox.ac.uk/`
- Twitter dataset. `https://www.kaggle.com/kazanova/sentiment140`
- Reddit dataset. This was collected from Reddit's API on 23rd March 2023. We cannot provide the data for copyright reasons. `https://www.reddit.com/.json`.

Code Availability

All code used to generate figures and analysis is available at `https://github.com/chasmani/PUBLIC-the-rising-entro`

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